

Welcome to CU Scarborough Project Planning

Project Lifecycle

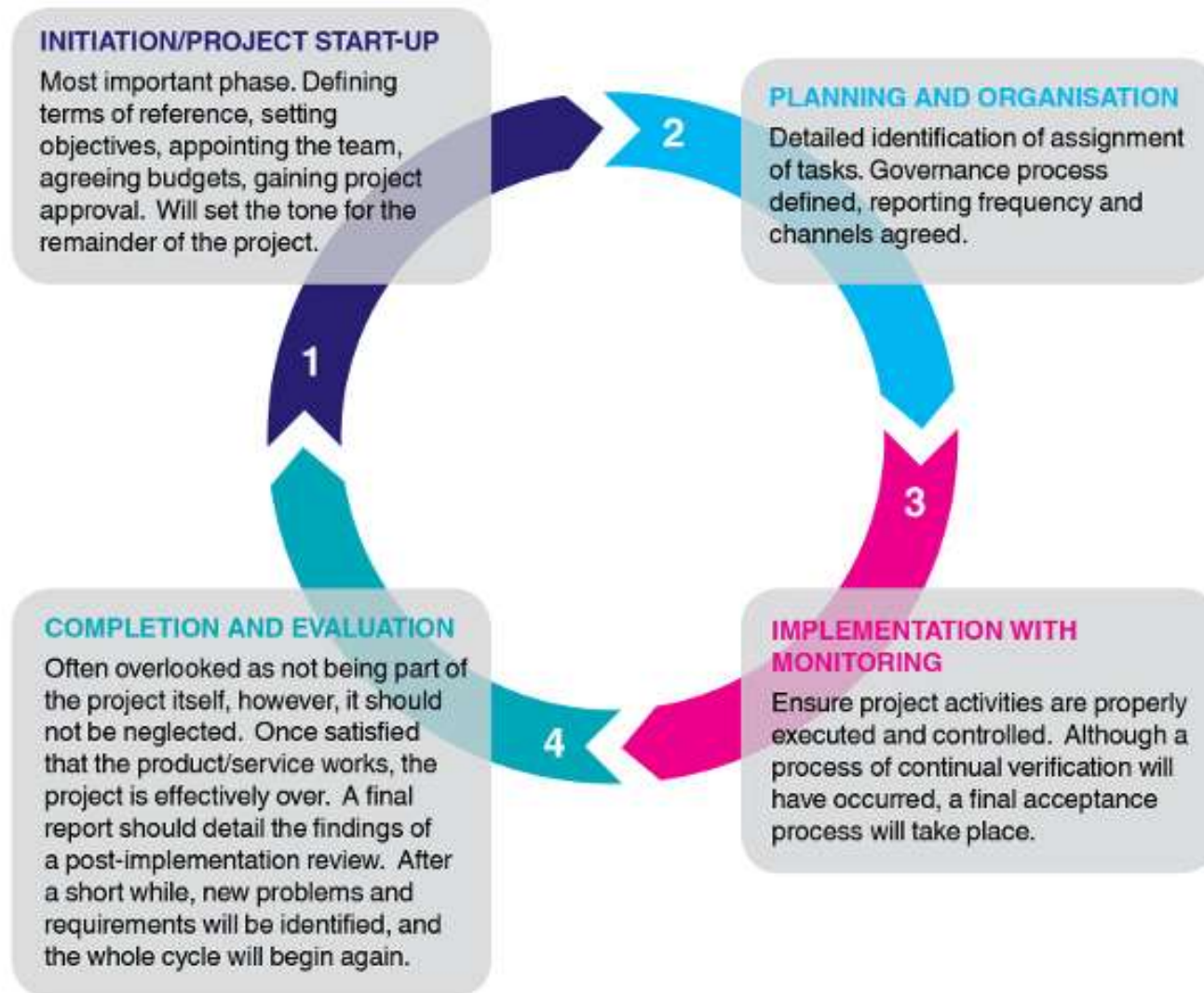
The **Project Life Cycle** refers to a logical sequence of activities to accomplish a project's goals or objectives.

Regardless of scope or complexity, any project will go through a series of stages during its life.

A project will typically begin as a result of a report or feasibility study which may have been run as an individual project, however, every project, whatever its size, should go through these stages:

- **Initiation**/project start up
- **Planning** and organisation
- **Implementation** with monitoring
- **Completion** and evaluation

Project Lifecycle



Project Lifecycle

Benefits:

By grouping activities into phases, the project manager and his team can efficiently plan and organise resources for each activity.

The manager is the driving force behind the project.

It is important to understand the implications of the project and to possess the managerial ability to co-ordinate the different elements.

Unless a project meets a genuine need, it is unlikely to succeed. A popular project management technique is PRINCE2.

Models - Elbeik and Thomas's successful project requirements

Concern about the high percentage of projects that were seen to be failing or exceeding time and budgetary limits led Sam Elbeik and Mark Thomas to explore the reasons for project failure.

They noted that in many cases team members were not sure of project objectives and came to the conclusion that time pressures on project managers meant that important parts of the project management process were ignored or neglected.

By speaking to experienced project managers they attempted to identify the critical factors that must be addressed if a project is to be completed successfully.

Based on the results of their success they also developed a six stage process for managing projects namely: define, plan, build the team, lead and motivate, control communications, review.

Benefits:

Many different tools and techniques are used for project management and it can be difficult to be clear about which are appropriate and when they should be used. The approach recommended by Elbeik and Thomas simplifies matters by identifying the critical issues that must be addressed. The ten factors provide a straightforward checklist of factors that should be taken into account when planning and carrying out a project.

Models - Elbeik and Thomas's successful project requirements

RANK	SUCCESS FACTORS
1	Clearly defined objectives
2	Good planning and control methods
3	Good quality of project managers
4	Good management support
5	Enough time and resources
6	Commitment by all
7	High user involvement
8	Good communication
9	Good project organisation and structure
10	Being able to stop a project

SMART GOALS in PROJECT MANAGEMENT

In project management, SMART goals are specific, measurable, achievable, relevant, and time-bound objectives that guide project planning and execution.

They ensure that teams are aligned and working toward achieving specific project outcomes within a set timeframe.

SMART GOALS



SMART GOALS in PROJECT MANAGEMENT

- Sometimes called [SMART objectives](#), SMART goals are used by project managers and leaders to create and communicate clear and measurable desired outcomes for people and projects.

These are the criteria for a SMART goal:

- **Specific:** The specifications are clear and exact.
- **Measurable:** The outcome can be quantified or measured.
- **Achievable:** The goal is realistic to achieve.
- **Relevant:** The goal relates to the goals of a particular project or the larger goals within an organization.
- **Time-Bound:** The goal includes a deadline or date by which success or failure will be determined.

SMART GOALS in PROJECT MANAGEMENT - EXAMPLES

- *“To improve communication with current, upcoming, and potential clients, I will investigate new collaborative software technologies. By implementing these tools, we can reduce communication delays and improve onboarding efforts. The measure of success will be a decrease in response time to clients within the next three months.”*
- *“To improve clarity, I will separate projects based on the clients and team members working on those projects. To achieve this, I will create new communication channels in existing collaborative tools and track communication responses. The goal is to create a channel for each major client within the month.”*
- *“To promote better communication and alleviate employee concerns, I will schedule weekly check-ins on current projects for one month to ensure we’re working within the schedule and budget constraints.”*

GANTT Chart

Developed by Henry Gantt, in 1917.

Provides a simple but powerful visual representation of a project plan by detailing how work is scheduled over time and how progress is monitored.

Gantt charts today are often generated using project management software but are also easy to construct using a spreadsheet. PERT (Program Evaluation and Review Technique) is a modern variation of the **Gantt chart**.

Bars or horizontal lines represent the beginning and end dates of each activity in the project.

Some bars may overlap denoting parallel tasks.

One task may have to be completed before another can begin and these dependency relationships can be indicated on the chart by simple links. Shading on the bars can indicate progress made on each task. Annotations can denote individuals or departments assigned to each task. Important events or milestones within the project can be marked on the chart.

GANTT Chart

Gantt Chart

Task Name	Q1 2019			Q2 2019		Q3 2019
	Jan 19	Feb 19	Mar 19	Apr 19	Jun 19	Jul 19
Planning						
Research						
Design						
Implementation						
Follow up						

GANTT Chart

Benefits of using the Gantt Chart:

The status of each activity and what has been achieved to date can be seen at a glance.

The chart keeps track of a project, illustrates how long a project will take and allows any necessary remedial action to be taken.

A simple Gantt chart can act as a communication tool to illustrate a project to senior management.

PERT Diagrams

PERT chart/diagram is a project management tool used to schedule, organise and coordinate tasks within a project.

It provides a graphical representation of a project's timeline that enables project managers to break down each individual task in the project for analysis.

The PERT chart template uses nodes -- drawn as rectangles or circles -- to represent events and milestones throughout the project. The nodes are connected by vectors -- drawn as lines -- which represent the various tasks that need to be completed.

PERT charts provide project managers with an estimation of the minimum amount of time needed to complete a project. Managers can also analyse the work breakdown and task connections, as well as assess the risk associated with the project. The breakdown structure makes it easy to organize a complex project with a variety of moving parts by visualizing the dependencies between each step of the process.

PERT Diagrams

How a PERT chart works

A PERT chart presents a graphic illustration of a project as a network diagram consisting of numbered nodes linked by labeled vectors. The direction of the arrows on the lines indicates the sequence of tasks.

When creating a PERT chart, project managers should follow the below steps:

1. Define all activities involved in the project.
 2. Consider any dependencies between tasks.
 3. Draw nodes and arrows based on the information gathered in the first two steps.
 4. Identify the completion time for each task.
- Once the PERT chart is drawn, project managers can use it to develop a realistic time frame for the project:
 - This can be done by finding the longest path, based on the estimations entered. This path should include the tasks that take the longest to complete.
 - By adding together the time it takes to complete each task, an estimation will be provided for how long the entire project will take.
 - Once this is done, the PERT chart can then be used to adjust the times if circumstances change or to ensure the project is finished by its designated deadline.

PERT Diagrams

When to use a PERT chart

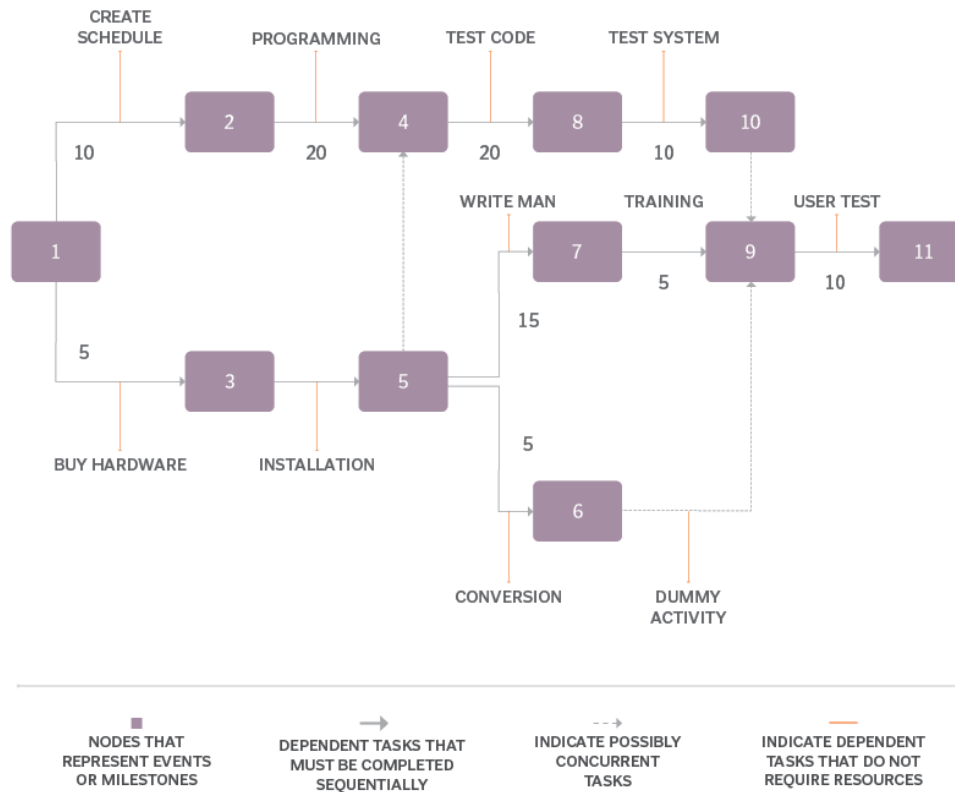
PERT charts should be used when a project manager needs to:

- determine the project's critical path in order to guarantee all deadlines are met;
- display the various interdependencies of tasks;
- estimate the amount of time needed to complete the project; and
- prepare for more complex and larger projects.

Project managers find best results when they use a PERT chart at the beginning of the project. Doing so helps ensure the project is accurately scoped and can help avoid any potential bottlenecks in the process.

PERT Diagrams

Program Evaluation Review Technique



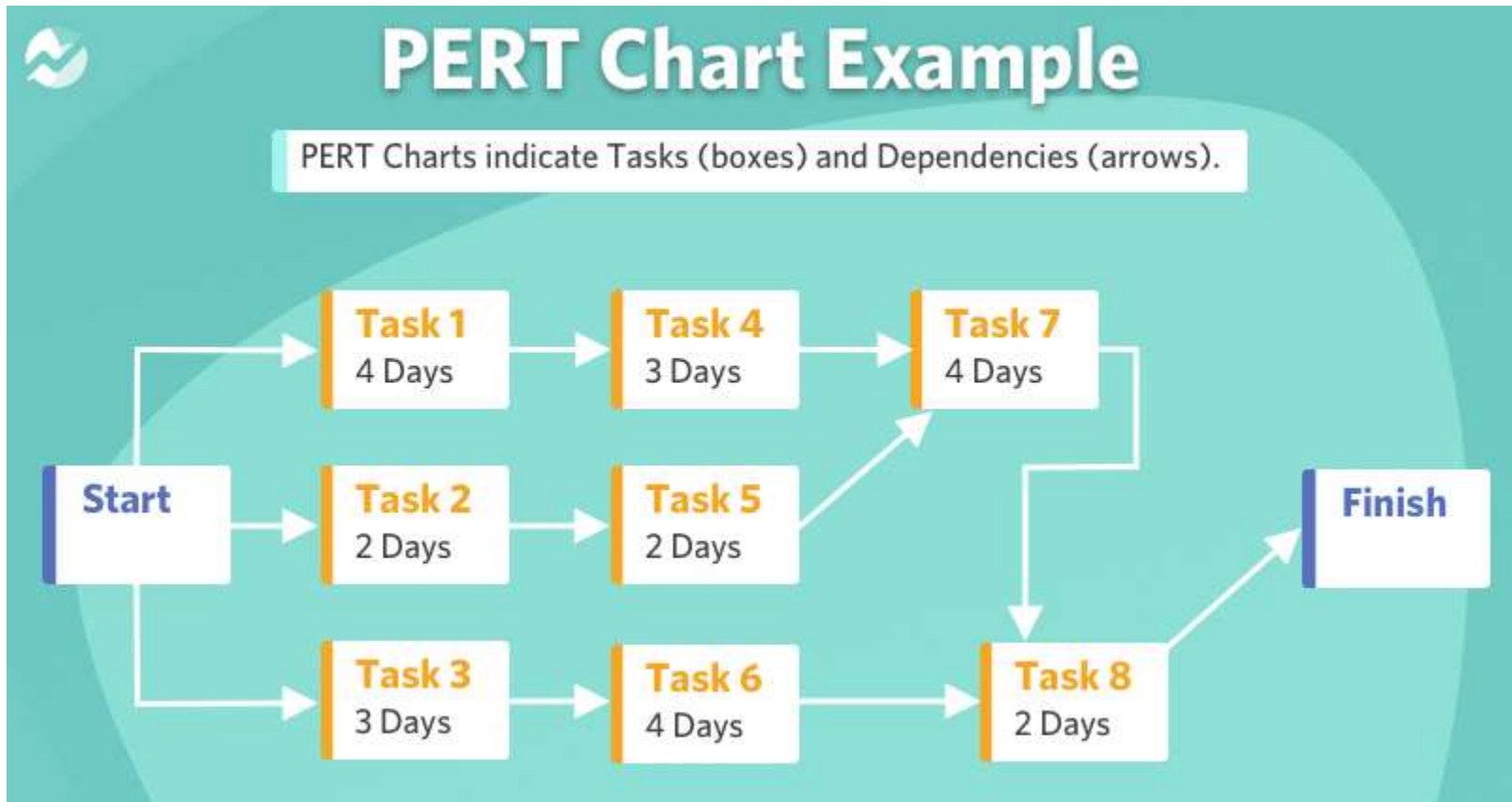
In the diagram, for example, the tasks between nodes 1, 2, 4, 8 and 10 must be completed in sequence. These are called *dependent* or *serial* tasks.

The tasks between nodes 1 and 2 and nodes 1 and 3 are not dependent on the completion of one to start the other and can be undertaken simultaneously. These tasks are called *parallel* or *concurrent* tasks.

Tasks that must be completed in sequence that don't require resources or completion time are considered to have event dependency. These are represented by dotted lines with arrows and are called *dummy activities*. For example, the dashed arrow linking nodes 6 and 9 indicates that the system files must be converted before the user test can take place but that the resources and time required to prepare for the user test -- writing the user manual and user training -- are on another path.

Numbers on the opposite sides of the vectors indicate the time allotted for the task.

PERT Diagrams



PERT Diagrams

Advantages of PERT charts

Project managers use PERT charts to gain these benefits:

- The PERT technique provides the ability to evaluate the time and resources necessary to a project by tracking required assets at each stage of the process, as well as throughout the course of the project.
- PERT charts are useful in what-if analyses, helping companies understand all possible [workflows](#) and choose the most efficient and beneficial path.
- The analysis of the PERT chart includes data from various departments within an organization. Combining all of the information helps identify each responsible team within the company, while facilitating an environment where each department takes responsibility for its work.
- The process of creating a PERT chart also improves communication and enables an organization to invest energy in projects that will enhance its strategic positioning.
- PERT charts make unclear [deadlines more predictable](#), clarify dependencies between tasks and establish a clear order for completing the tasks.

PERT Diagrams

Disadvantages of PERT charts

Disadvantages of the PERT chart include the following:

- A strict focus on deadlines may not enable managers to see the full financial positioning of the project.
- PERT charts lack the flexibility to adapt to small changes that occur when confronted with a roadblock.
- If any calculations are inaccurate in the creation of the chart, delays could occur, causing bottlenecks and negatively impacting the final delivery date.
- PERT charts are subjective; their success depends on the experience of the project manager. Consequently, some charts may include unreliable data or unrealistic expectations for the cost and time frame of the project.
- Creating a PERT chart is labour-intensive, requiring additional time and resources. Also, in order for the chart to remain valuable, it must be consistently reviewed and maintained.

Models - MoSCoW Prioritisation



MUST have this – essential requirement as key stakeholder needs will not be satisfied if this requirement is not delivered, and the timebox will be considered to have failed



SHOULD have this if at all possible – an important requirement, but if it is not delivered within the current timebox, there is an acceptable workaround until it is delivered during a subsequent timebox



COULD have this if it does not affect anything else – a 'nice to have' requirement; we have estimated that it is possible to deliver this in the given time but will be one of the requirements re-scoped if we have underestimated.



WON'T have this time but WOULD like in the future - the full name of this category is 'Would like to have but won't have during this timebox'. Requirements in this category will not be delivered within the timebox that the prioritisation applies to.

Models - MoSCoW Prioritisation

MoSCoW* is a semantic project management prioritisation technique which uses words rather than numbers to reach an understanding with stakeholders on the importance that is placed upon project requirements and their priority.

Successful projects require a prioritisation of the requirements and the main project objectives. To do this, a method with semantic value such as MoSCoW is a useful technique to use. The MoSCoW method (first developed by Dai Clegg of Oracle UK Consulting) is often used with timeboxing, where a deadline is fixed so that the focus can be on the most important requirements.

Project success often depends on ensuring that requirements are prioritised.

It is essential that a clear set of prioritised requirements are agreed with the internal or external customer along with the overall objective, quality, timescale and budget.

Models - MoSCoW Prioritisation

A more useful method can be to use a set of words that have meaning such as the MoSCoW method. Using words can make it easier to discuss what is important.

The 'must' requirements need to provide a complete coherent solution which will alone lead to project success. These requirements are non-negotiable and have to be delivered.

Failure to deliver even one of these essential requirements means the project has failed.

The project team should aim to deliver as many 'should' requirements as possible. 'Could' and 'would' requirements are "nice to haves" and do not affect the overall success of the project. – they are the first to go if the project timeline or budget comes under pressure.

Reading resources



Required Reading:

- Larson, E. and Gray, C. (2014) *Project Management: The Managerial Process*. 6th edn. New York: McGraw-Hill
- Lock, D. (2013) *Project Management*. 10th edn. Farnham: Gower
- Project Management Institute (2000) *A guide to the Project Management Body of Knowledge (PMBOK Guide)* 2000 edn. Pennsylvania

Recommended Reading:

- Kerzner, H. (2017) *Project Management: A Systems Approach to Planning, Scheduling, and Controlling*. 12th edn. Hoboken: Wiley
- Newton, R. (2009) *Project Manager: Mastering the Art of Delivery*. 2nd edn. Harlow: Pearson
- Westland, J. (2007) *The Project Management Life Cycle*. London: Kogan Page
- Wysocki, R. (2014) *Effective Project Management: Traditional, Agile, Extreme*. 7th edn. Hoboken: Wiley
- CMI Website : Management Direct